

$$\begin{aligned}
P_{MC}(0,1) &= \underbrace{P_{C|M}(1|0)} P_M(0) \\
&= IP[M \oplus K = 1 \mid M=0] \\
&= IP[K=1 \mid M=0] \\
&= IP[K=1] \quad \text{because } K \text{ and } M \text{ are independent} \\
&= \frac{1}{2} \\
&= P_C(1)
\end{aligned}$$

Generalization to a message M of length n bits

Encryption: $C = M \oplus K$, where $M \in \{0,1\}^n$

$$\forall k \in \{0,1\}^n, IP[K=k] = \frac{1}{2^n}$$

- $K \in \{0,1\}^n$ is uniformly distributed and independent of M
- $\oplus$ is the bitwise modulo 2 addition

Example

$$n=5$$

$$M = 01011$$

$$K = 11011$$

$$C = 10000$$

Decryption: Compute $C \oplus K = \underbrace{(M \oplus K) \oplus K}_{=0} = M$

Security analysis: We want to prove that
 M and C are independent, i.e.
 $P_{MC} = P_M \times P_C$, i.e.

$$\forall m \in \{0,1\}^n, \forall c \in \{0,1\}^n, P_{MC}(m,c) = P_M(m) \times P_C(c), \text{ i.e.}$$

$$\forall m \in \{0,1\}^n, \forall c \in \{0,1\}^n, P_{MC}(m,c) \times P_C(c) = P_M(m) \times P_C(c)$$

Proof: Fix $m \in \{0,1\}^n$ and $c \in \{0,1\}^n$

$$\text{We have } P_{MC}(m,c) = \frac{P_{c|m}(c|m) P_M(m)}{P_C(c)} \quad \begin{matrix} \text{(Bayes' rule)} \\ (1) \end{matrix}$$

$$\begin{aligned} \text{We have } P_{c|m}(c|m) &= \mathbb{P}[C=c \mid M=m] \\ &= \mathbb{P}[M \oplus K = c \mid M=m] \\ &= \mathbb{P}[m \oplus K = c \mid M=m] \\ &= \mathbb{P}[m \oplus m \oplus K = m \oplus c \mid M=m] \\ &= \mathbb{P}[K = m \oplus c \mid M=m] \\ &= \mathbb{P}[K = m \oplus c] \quad \text{by independence} \\ &= \frac{1}{2^n} = 2^{-n} \quad \text{between } M \text{ and } K \end{aligned} \quad \begin{matrix} (2) \end{matrix}$$

$$\begin{aligned} \text{We have } P_C(c) &= \sum_{m'} \underbrace{P_{c|m'}(c|m')}_{= 2^{-n}} P_M(m') \quad \begin{matrix} \text{(by the law} \\ \text{of total probability)} \end{matrix} \\ &= 2^{-n} \sum_{m'} P_M(m') \\ &= 2^{-n} \end{aligned} \quad (3)$$

Hence, by (1), (2), and (3), we have

$$P_{M1C}(m1c) = P_M(m)$$

and thus $P_{M1C}(m1c) \times P_C(c) = P_M(m) \times P_C(c)$, i.e.,

$$P_{MC}(m1c) = P_M(m) \times P_C(c)$$

Limitations

- How to exchange the secret key?
- You can only use the key once.

$$C_1 = K \oplus M_1$$

$$C_2 = K \oplus M_2$$

An adversary can compute $C_1 \oplus C_2$

$$= K \oplus M_1 \oplus K \oplus M_2$$

$$= M_1 \oplus M_2$$

- The key must be as long as the message

(we lose independency between the ciphertext and plaintext)

• Proof that the length of the Key must be as long as the message M .

By contradiction,

let $\{0,1\}^{m_1}$ be the set of values that the key K can take

let $\{0,1\}^{m_2}$ be the set of values that the message M can take

assume that $m_1 < m_2$, and for any $m \in \{0,1\}^{m_2}$,
 $IP[M=m] \neq 0$